

Let's start from Switching Attacks. The goal of this kind of attacks is to bring the power system to a state of instability. Attackers perform this attack by acting on switches, and that's why they take this name.

Switches are the interrupters we have in power systems, and we can have line switches. They're placed along lines, to connect and disconnect them. There are also load switches, which are used to connect and disconnect a load. We also have generator switches. The goal is thus to cause instability on the network by acting on the switches. Connecting or disconnecting a component will indeed cause a perturbation, and the goal for us is to understand possible ways of acting on switch.

We focus firstly on load switches. Acting on switches means to find a right switching sequence, which will cause a serious perturbation on the grid.

Instability basically means that one of the variables goes to ∞ .

We take the grid, and specifically a target on which the attacker focuses, like a generator. The goal of the attacker is to cause the disconnection of the generator. Close to the generator we have a load, connected to the line with a switch.

This attack is actually a simplified scenario.

The grid, on which we do not focus, could be modelled as an infinite bus generator.

We don't act on the switch of the generator itself, rather on the switch of the load, this will lead to repercussions to the rest of the grid, including the targeted generator.

We focus on single switching attacks. The generator is modelled with the swing equations. They are given by the following:

$$\begin{cases} \dot{\delta} = \omega \\ \dot{\omega} = \dots \text{balance of power} \end{cases}$$

The state variables of the generator are given by (δ, ω) . They must remain close to a set point, and when they go too far away, the machine disconnects. The goal of the attacker is indeed to bring these variables out of the safety limits, typically the value of δ .

So, $x = \begin{bmatrix} \delta \\ \omega \end{bmatrix}$ of the target generator.

Now, let's model mathematically the scenario and the attack. We'll then try to identify a possible switching sequence used to destabilize the system.

We don't need the actual equations, rather we're going to work with so-called switched system model.

We define s as the status of the switch of the load. Yesterday we modelled s as a binary variable, thus $s : \{0, 1\}$. Today, we say that $s = \begin{cases} > 0 & \text{switch is closed} \\ \leq 0 & \text{switch is open} \end{cases}$

In general, a dynamical system can be modelled, in the state space, as $\dot{x} = f(x)$.

We ignore inputs for now.

Here, we have:

$$\dot{x} = \begin{cases} f_1(x) & \text{if } s > 0 \\ f_2(x) & \text{if } s \leq 0 \end{cases}$$

So, we have two models: one when the load switch is open, and one when it's closed.

The goal is thus to find a switching sequence, with $s(t)$ (so, s a function of time), to make the system unstable. The attacker will take control of the switch, and change the status of it over time.

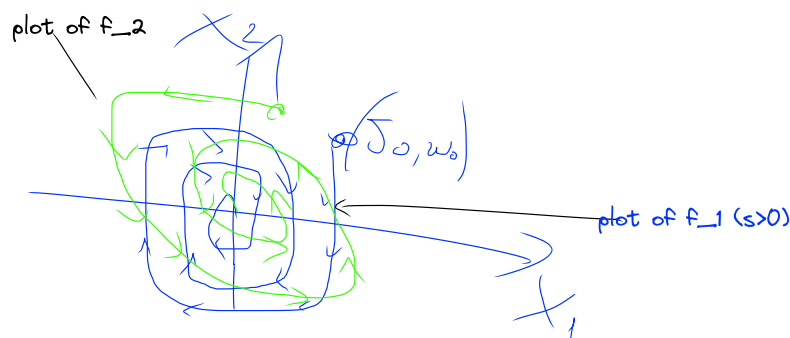
The attacker has to find a destabilizing switching sequence. There are many ways to do this, but we take a look at one in particular.

One notable fact beforehand: Consider the two models separately, $\dot{x} = f_1(x)$ and $\dot{x} = f_2(x)$. Normally, they're typically stable systems. Even when working with two stable systems, when cleverly switching we could get unstable behaviours. Random switching sequences will not destabilise the system.

We focus now on a practical approach to do this.

We write the phase portrait of the system (for f_1 and f_2). This is a graph in which we plot f_1 and f_2 , given the axis given by x_1 and x_2 .

Imagine we start from an initial value of (δ_0, ω_0) .



An attacker writes the switching sequence $s(t)$ as a function of the state x . We'll write thus $s(x)$. Can we find a function that makes the switched system unstable?

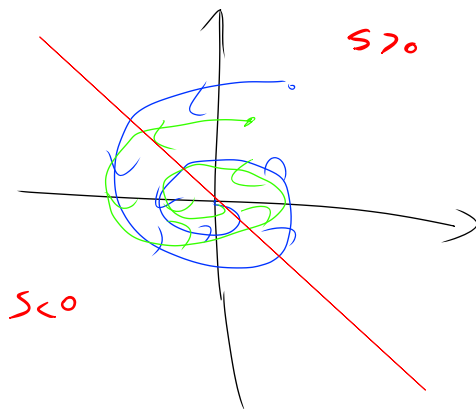
The attacker will measure state variables $x = \begin{pmatrix} \delta \\ \omega \end{pmatrix}$ and he will compute the value

of $s(x)$. Based on the value of s , the attacker will change the status of the switch. It's a feedback attack as the attacker needs disclosure resources.

The core problem is to find the right $s(x) = s(x_1, x_2)$ function. We can start with a linear function, given by $s(x) = c_1x_1 + c_2x_2$. We need to find thus the values of c_1 and c_2 . We define $s(x) = 0$ as the switching surface, as when the state crosses the surface, the switch changes status.

We need thus $c_1x_1 + c_2x_2 = 0$, and find the correct line in the graph.

Consider we have two modes



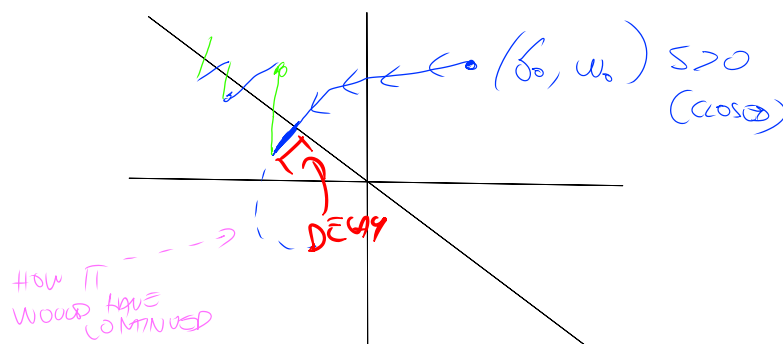
If we take $c_1 = c_2 = 1$, we have that $x_1 + x_2 = 0$, and thus $x_2 = -x_1$. *Bisettrice del secondo e quarto quadrante*. If we take any point above the line, the value of s will be greater than zero, lower than zero when the point is below instead.

Only when we cross the line we have a change in the status.



Let's say the attacker picks this switching surface.

We consider an initial state of the system (δ_0, ω_0) . We're above the switching surface.



After passing the surface, we'll have a little delay in the system (so, from the point of view of the system, the switch is still closed). After the small delay, the behaviour will change. The system will go up, so the switch will be closed again, and after a delay the system will keep infinitely to change between the open and closed status.

This specific mode of changing takes the name of unstable sliding mode.

The key is to find a linear $s(x)$ resulting in an unstable sliding mode. What do we mean with the keywords?

- Unstable $\rightarrow$ the trajectories of the switching system need to go away from the origin.
- Sliding mode $\rightarrow$ the trajectories "slide" along the line given by $s(x) = 0$, as the trajectories will remain close to $s(x) = 0$.

The construction of s can be found from the graphical construction. With the keyword sliding, we mean that the two modes ($s > 0$, $s < 0$) point towards $s(x) = 0$.

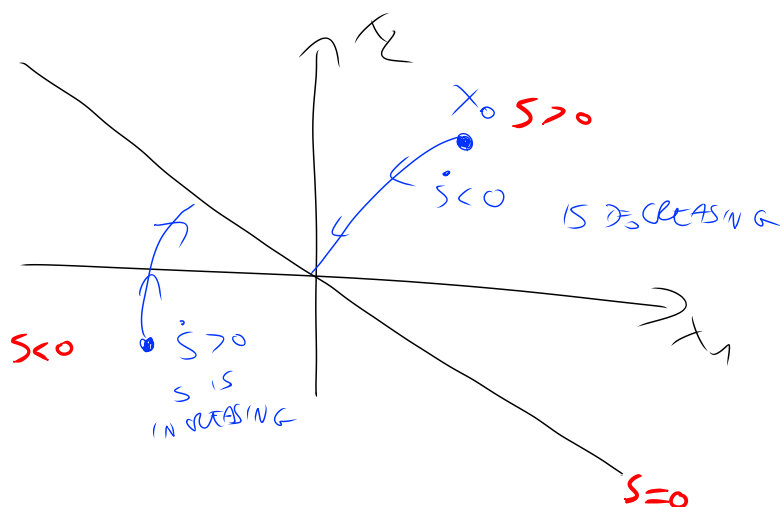
For unstable, they point/go away from the origin.

For a formal approach, instead, we need two necessary and sufficient conditions necessary to describe sliding and unstable features. We focus only on sliding.

The mathematical condition to check if $s(x) = 0$ is a sliding surface.

$$s(x) \cdot \dot{s}(x) < 0 \quad \forall x : s(x) \neq 0$$

If we're able to find a $s = c_1x_1 + c_2x_2$ such that $s(x) \cdot \dot{s}(x) < 0$, then $s(x) = 0$ is sliding/attractive.



With a negative derivative, the trajectory goes down, moving closer to points in which $s = 0$. Similar to when we start from below the line, in that case the derivative is positive.

We also need another condition to impose an unstable sliding mode (*but we don't look at this*).

A final consideration: this attack focuses on a single switch, the attacker requires disclosure and disruption resources, as well as model knowledge (we need to know the equations of the system).

A more realistic attack is given by the use of several switches controlled, and the attacker might want to minimise the number of switching needed to bring δ, ω out of safety limits. He may also want to respect the constraint of an actuator (switch), as well as minimising the attack time, so the time needed to bring x outside of the safety limits. In reality, the model is non linear, having a more complex design. We saw the linear approach, so that $\dot{x} = f(x)$ is linearized, and f_1 and f_2 are linear functions. We thus have:

$$\dot{x} = \begin{cases} {}_1x & s > 0 \\ {}_2x & s \leq 0 \end{cases} \quad \text{linear model of the grid}$$

$$s(x) = c_1x_1 + c_2x_2 \quad \begin{array}{l} \text{linear switching} \\ \text{rule} \end{array}$$

Note: $\dot{s} = \frac{1}{t}s(x(t)) = \frac{1}{x}s \cdot \frac{x}{t}$, showing how it depends on the model of the system. Using these formulas, we can find conditions on c_1 and c_2 to make the sliding mode attractive.

Advanced attacks try to avoid using too much model knowledge, consider a non linear system, use multiple switches, and use a non linear s , as we have less switching and a faster attack.

Load Altering Attacks

They are a generalisation of load switching attacks, as here the attacker takes control of some loads and alters the power consumption in a coordinate way, to destabilise the grid (typically the grid frequency). The practical effects are given by the disconnection of a generator.

The scheme is similar to the one seen a lot of times. The attacker will need a frequency meter placed somewhere in the grid, which will yield information about $\omega(t)$. This information is then given to a particular attack logic module, which will send a command to the loads under control. The attack will change in time the amount of power that the load is consuming. We have a target bus/generator (the

attacker wants to disconnect this generator), and we call it v . The attacker wants to bring ω_v outside of safety limits, by dynamically controlling the power consumption at victim's loads.

Generally, we need to take control of several loads.

Imagine a scenario in which the attacker takes control of a load. We can have several types of load altering attacks, like:

- Single point/multi point (we could control one or several loads)
- Static/dynamic: Imagine an attacker is able to take control of a load. A static attack will change the consumption of the load at a specified level (like the maximum level). This is a naive attack, and are not problematic for the grid, which will be able to recover easily. A dynamic attack, instead, sees the alteration of the load changing in time, according to pre-defined formulas.
- Open loop/closed loop: For the open loop, the load alteration is not computed as a function of the current state, and is pre-computed. For a closed loop, instead, the load alteration is computed based on measurement of frequency. A closed loop attack might result in a stable system, so not a successful attack, or an unstable system (successful).

We're going to focus on closed-loop dynamical load altering attacks.

For the model of the system, we'll use the grid modelled with power flow equations, and generators modelled with swing equations. Putting together these things we get a descriptor system, as we're combining algebraic equations (power flow eq.) with differential equations.

We start by modelling the attack. We need to decide:

- The attack law: The power we change is a function of frequency, and so we need to find this function. We start from intuitive linear function.
- Which loads to attack, so on which buses of the network the attack has to be deployed, as well as how large a portion of the load on each node do we need to control.

The criterion is to check stability/unstability of the resulting attack. This is the Lyapunov criterion.

The setting is this: We have a power system with a given number of buses. Some of them carry load, while others carry generators. We have a sensor deployed somewhere in the grid. We work with linearized power flow equations. These are the node equations, as they tell us the power consumed in a given load. We have the power generated P^G , and the power consumed by loads P^L .

The voltage angle at the generator bus is denominated δ , and for the load bus is called θ . We then have the equations for the generators. Here δ is the voltage angle, and ω the angular velocity. The machine is pushed by the mechanical power P^M , and P^G is the generated power which slows down the machine, and we have friction. Let's focus on P^M and P^G . The former is the one controlled by the Prime Governor Controller, which adjusts the mechanical power to keep ω and δ at the reference. If the controller sees the machine is accelerating, it will remove some mechanical power. We're going to model the controller, and the typical one is the PI (proportional integral) controller of P_i^M .

The controller will compute P^M (the variation of the mechanical power) as two components:

- One proportional to ω
- One proportional to the integral of ω

Let's focus on the first component, $P_i^M = -K_i^P \omega_i$.

When $\omega_i > 0$, the deviation is positive, and so the machine is spinning too fast. Thus, we have too much mechanical power compared to the currently requested power.

The minus has a stabilising effect. When instead $\omega_i < 0$, we have demand from the grid, and we don't have enough mechanical power. So, $P_i^M > 0$, which means we need to increase the mechanical power.

Let's now focus on P^G . It's the power generated by the generator, but also the result of the balance of consumption across the grid. Even if the generators are often well-protected, they still depend on the rest of the network.

As for the attack law, we'll have that $P_i^G = -K\omega$. If we have a positive gap, thus have more power generation than consumption, the attacker will further increase the gap.